



Taxi Fare Meter

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A taxi meter in cabs and autorickshaws calculates the fare based on the distance travelled. Here is a circuit that calculates the trip fare based on the rate per km and displays the distance travelled and the total fare.

The circuit (see Fig. 1) comprises Arduino Uno Board1, 16 × 2 LCD display LCD1, and inductive proximity sensor connected across CON2. A metal target such as a bolt-head required by the sensor for proper functioning is fixed on the rotating object such as a wheel's rim or a wheelnut to

sense the rotation.

Whenever the metal target crosses the sensor while rotating, the sensor generates a pulse. The pulse goes to the first interrupt pin (pin 2) of the Arduino through diode D1. The diode acts as a voltage level converter. Since the proximity sensor's working voltage is 12V, its output pulse voltage is 0-12V.

The Arduino cannot withstand 12V voltage, so the diode is used to change it to 5V logic pulse for the Arduino. If the sensor output is high, the diode does not conduct and the 5V is available at pin 2 of the Arduino through

pull-up resistor R2. If the sensor output is low, the diode gets forward biased and pulls pin 2 to ground, thus providing logic low at pin 2 of Board1.

The Arduino counts the number of pulses for one rotation of the wheel and converts to distance in kilometre (km) through the program. The distance is then multiplied with the rate per km to get trip fare for the distance travelled. LCD1 displays the distance and trip fare.

Before the vehicle is hired, the LCD1 displays the message "TRIP IS TO BEGIN, HAPPY JOURNEY."

When it gets hired, switch S1 is pressed and LCD1 now displays "TRIP STARTED, HAPPY JOURNEY" for half a second. Thereafter, the distance travelled and the trip cost is displayed on LCD1 continuously.

On completion of the trip, the total fare displayed on LCD1 is noted and switch S1 is pressed again to register completion of the trip. Now LCD1 displays "TRIP COMPLETED, THANK YOU" for one second and then displays the first message again for the next trip.

PNP type normally-open (NO) proximity sensor used in the circuit (see Fig. 2) detects objects without any physical contact up to its nominal range—the maximum distance

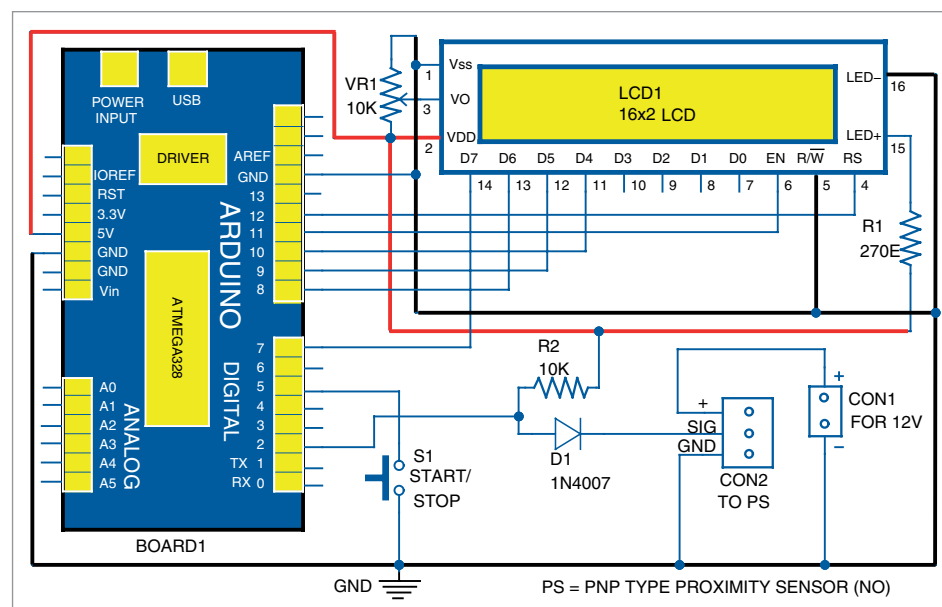


Fig. 1: Circuit diagram



Fig. 2: Proximity sensor

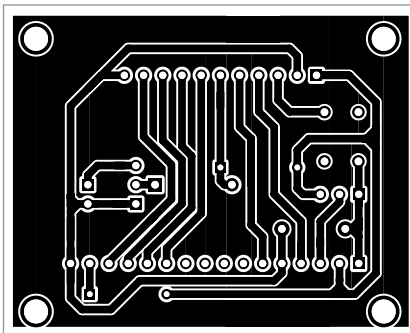


Fig. 3: An actual-size PCB layout

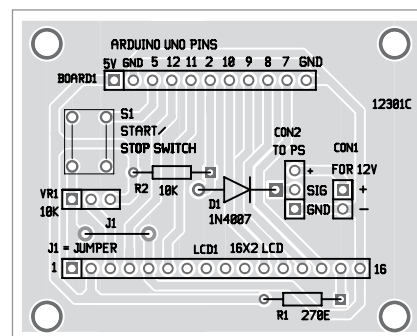


Fig. 4: Component layout